

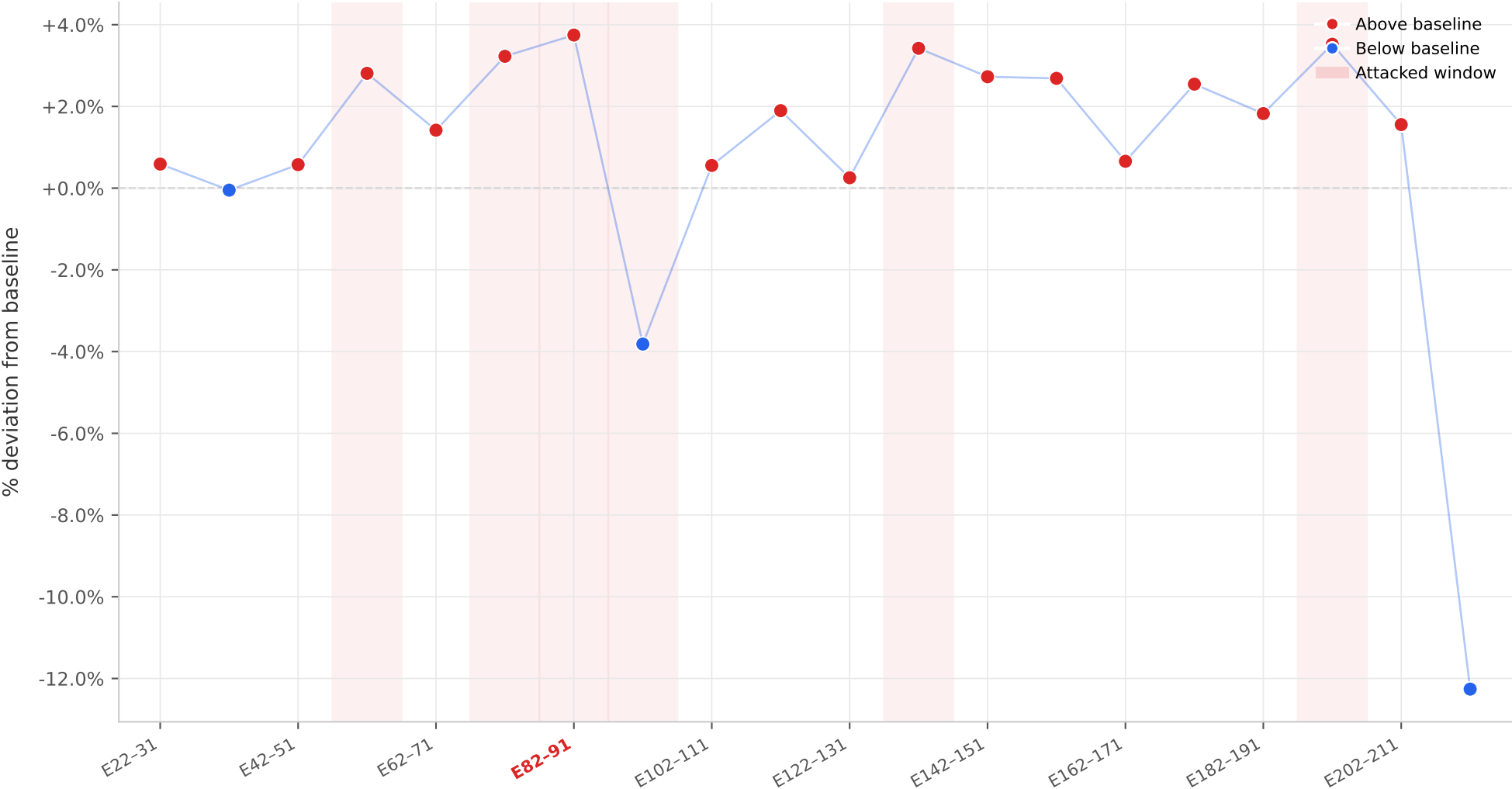
Entropy Stream Analysis

Deviation from Baseline — Per-Metric Report

Run Parameters		Baseline Values	
Dataset	txtD280_10.0_2.0_sparse.json	Src IP Entropy	40.87
Directory	CAIDA_Dataset/txtD280	Dst IP Entropy	32.31
Timestamp	2026-02-18T16:07:20.997483	Src Port Entropy	21.73
Total epochs	215	Dst Port Entropy	43.29
Baseline epochs	21 (10.0%)	Src IP Cardinality	5045
Attack epochs	194	Dst IP Cardinality	5027
Malicious traffic	2.0%	Src Port Cardinality	4302
Epoch group size	10 epochs per point	Dst Port Cardinality	5042
Metrics plotted	16	Src IP Uniformity	4.292e+08
Mode	SPARSE	Dst IP Uniformity	1.236e+06
Attack probability	0.3	Src/Dst Ratio	1.003
Windows total	20	Total Flows	5078
Windows attacked	6	Unique Src IPs	3579
Windows clean	14	Unique Dst IPs	2804
		Unique Src Ports	2323
		Unique Dst Ports	3825

Source IP Entropy

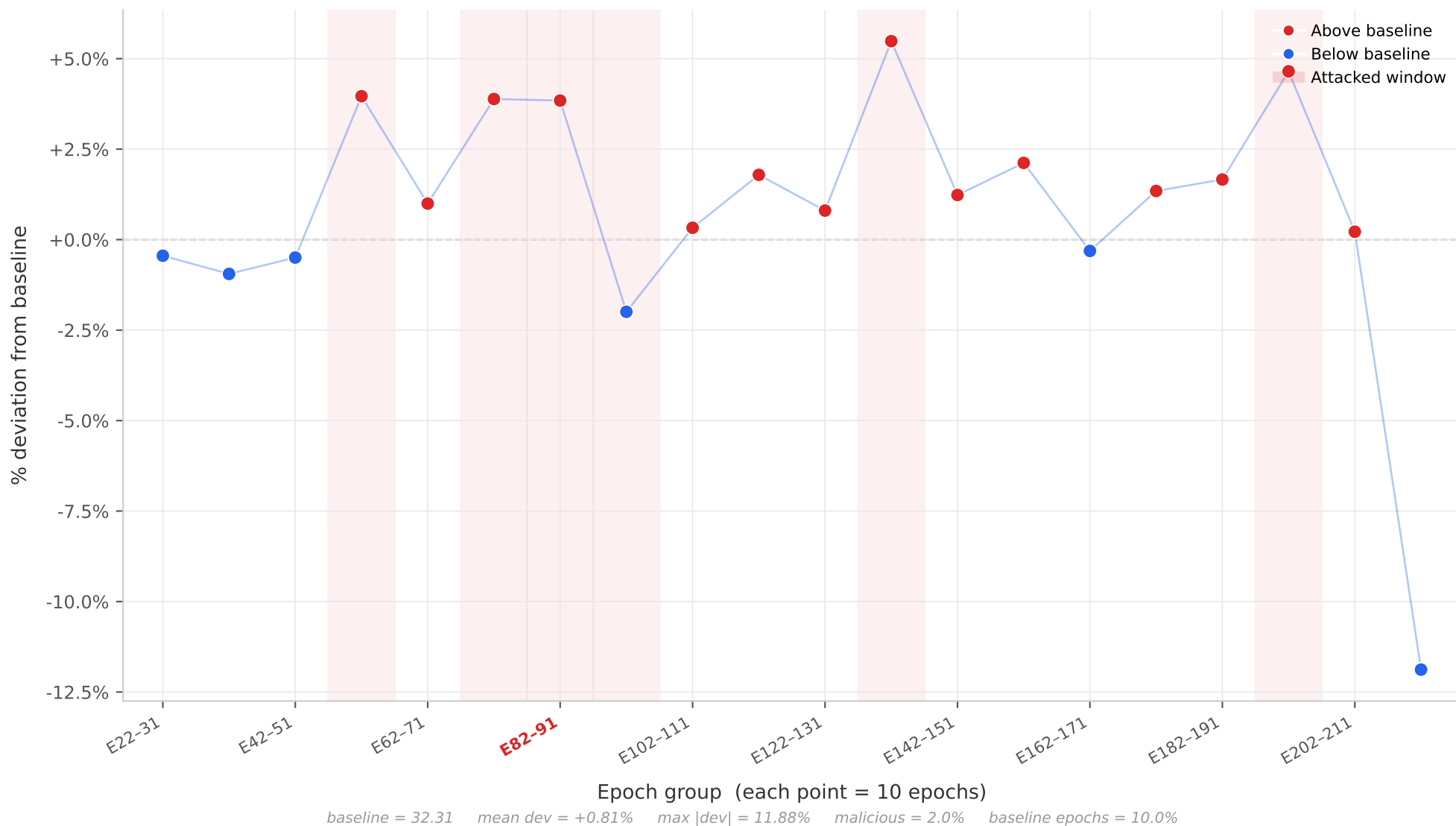
Entropy



baseline = 40.87 mean dev = +0.89% max |dev| = 12.26% malicious = 2.0% baseline epochs = 10.0%

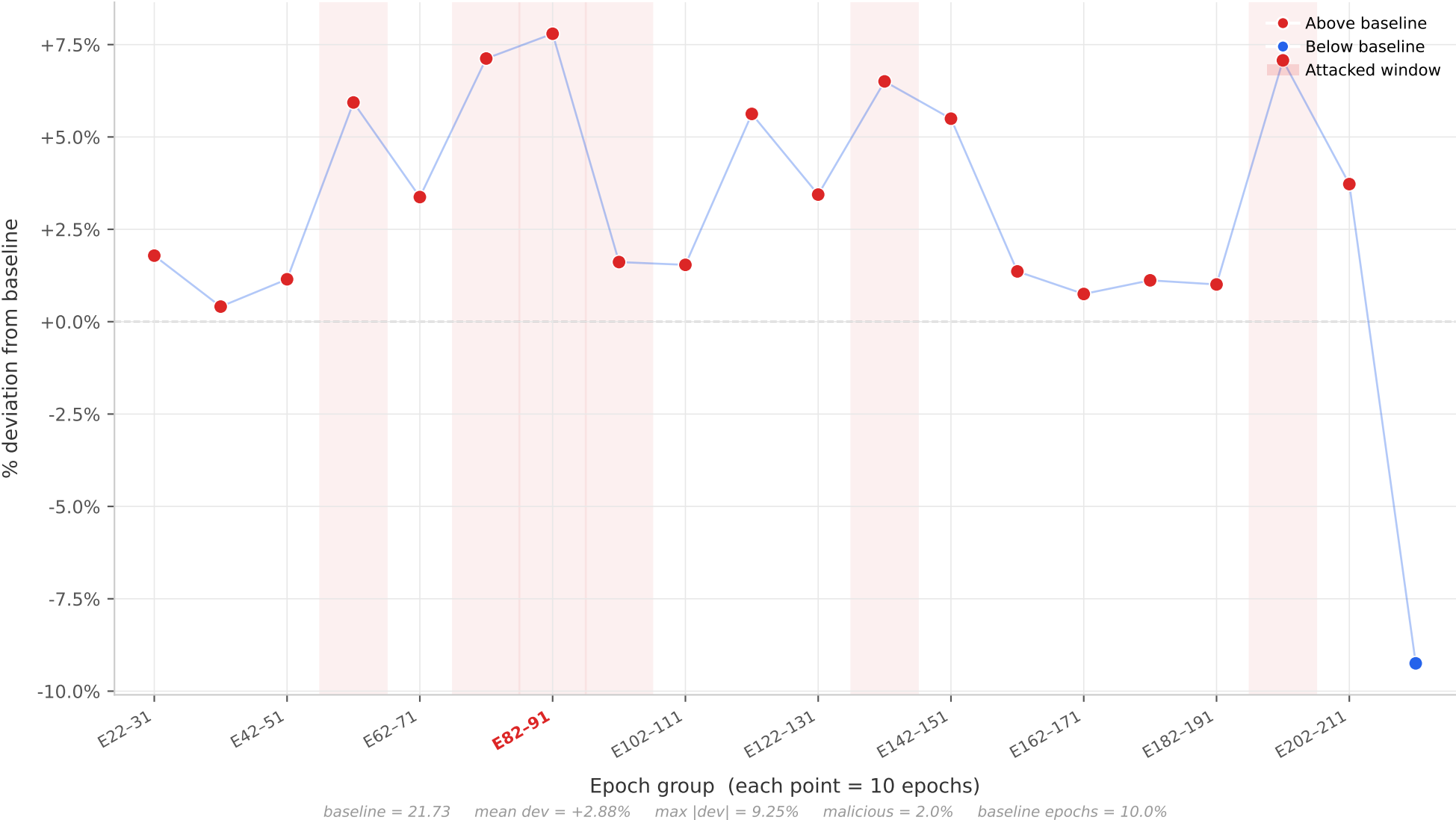
Destination IP Entropy

Entropy



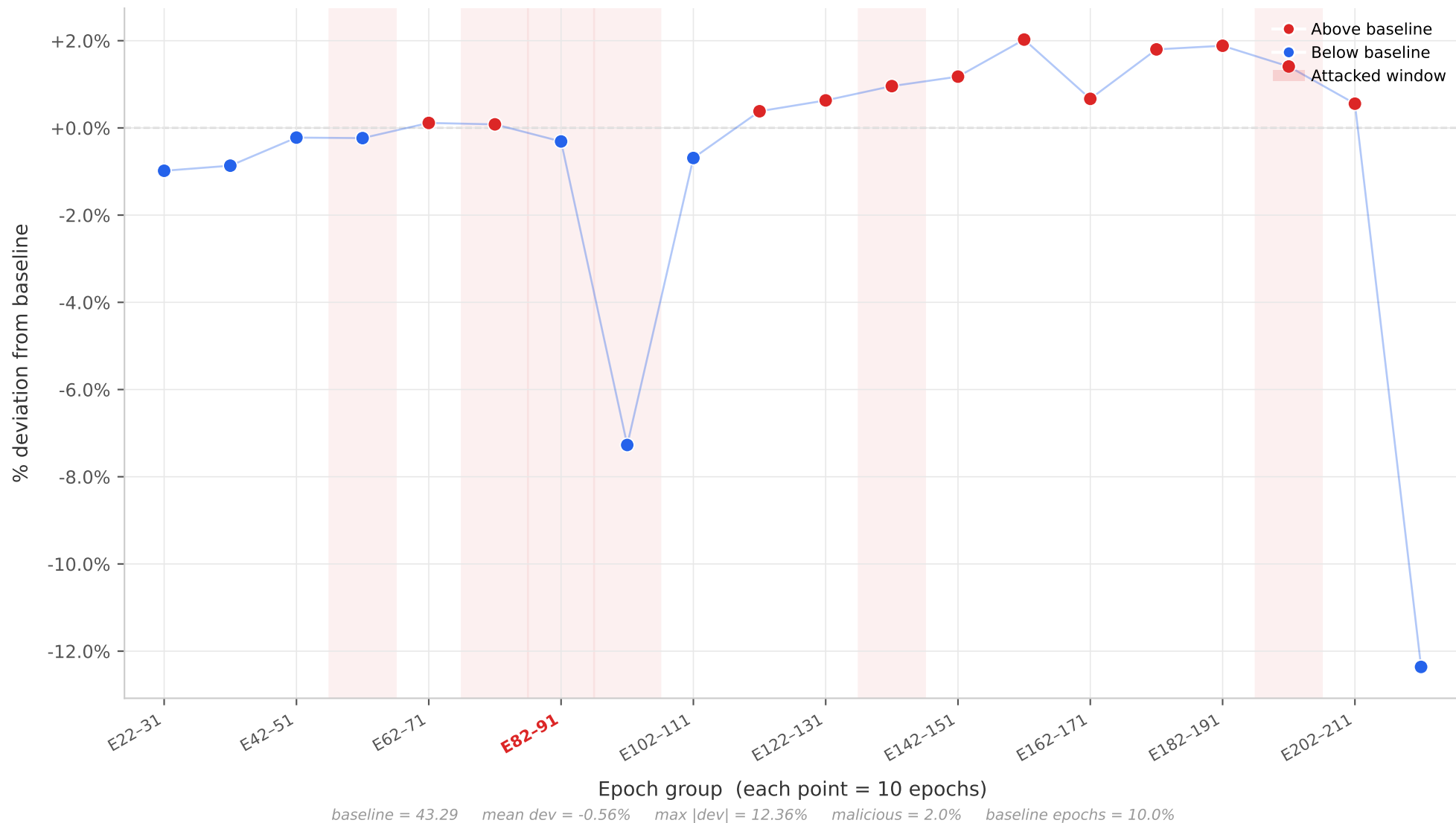
Source Port Entropy

Entropy



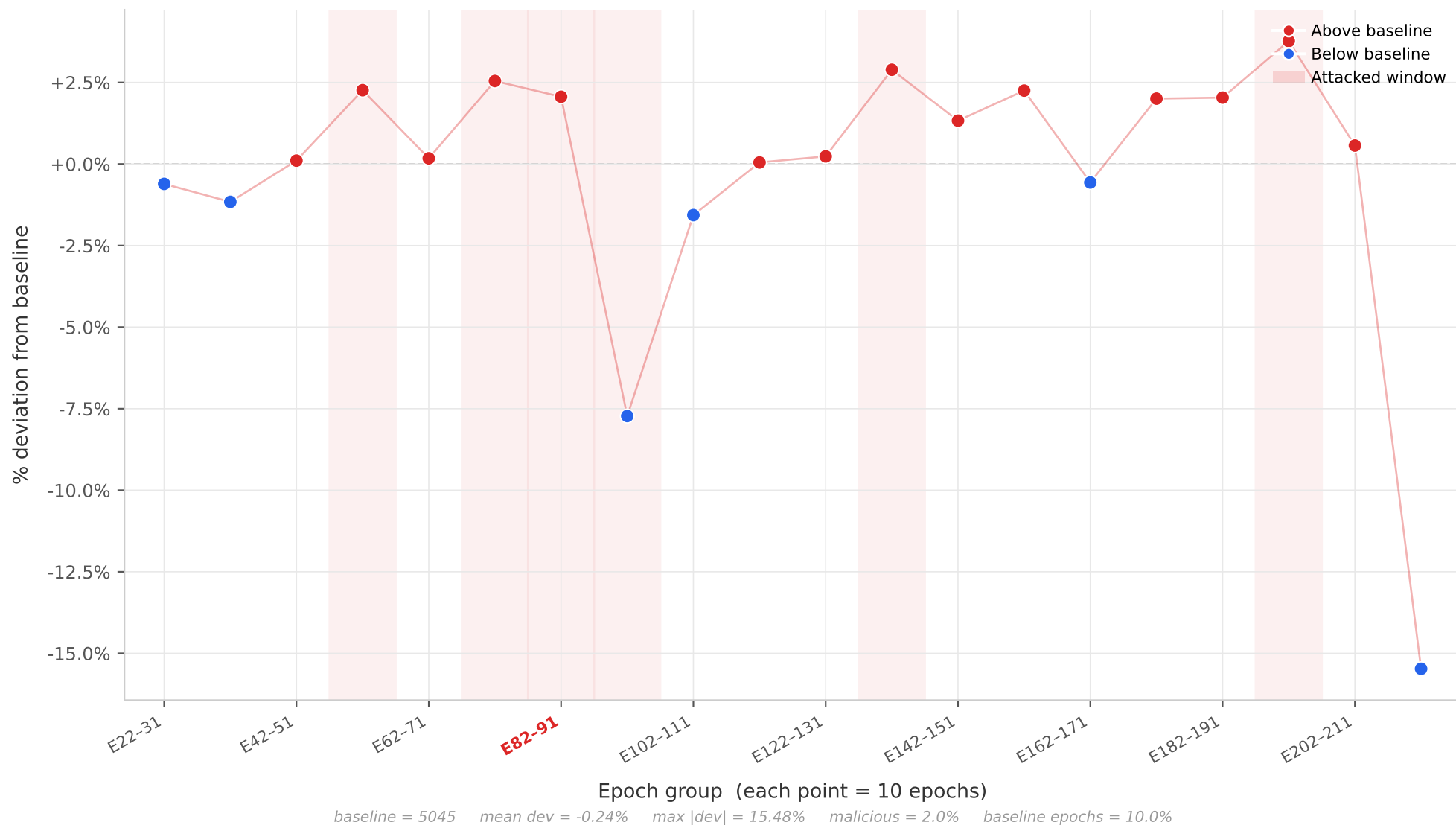
Destination Port Entropy

Entropy



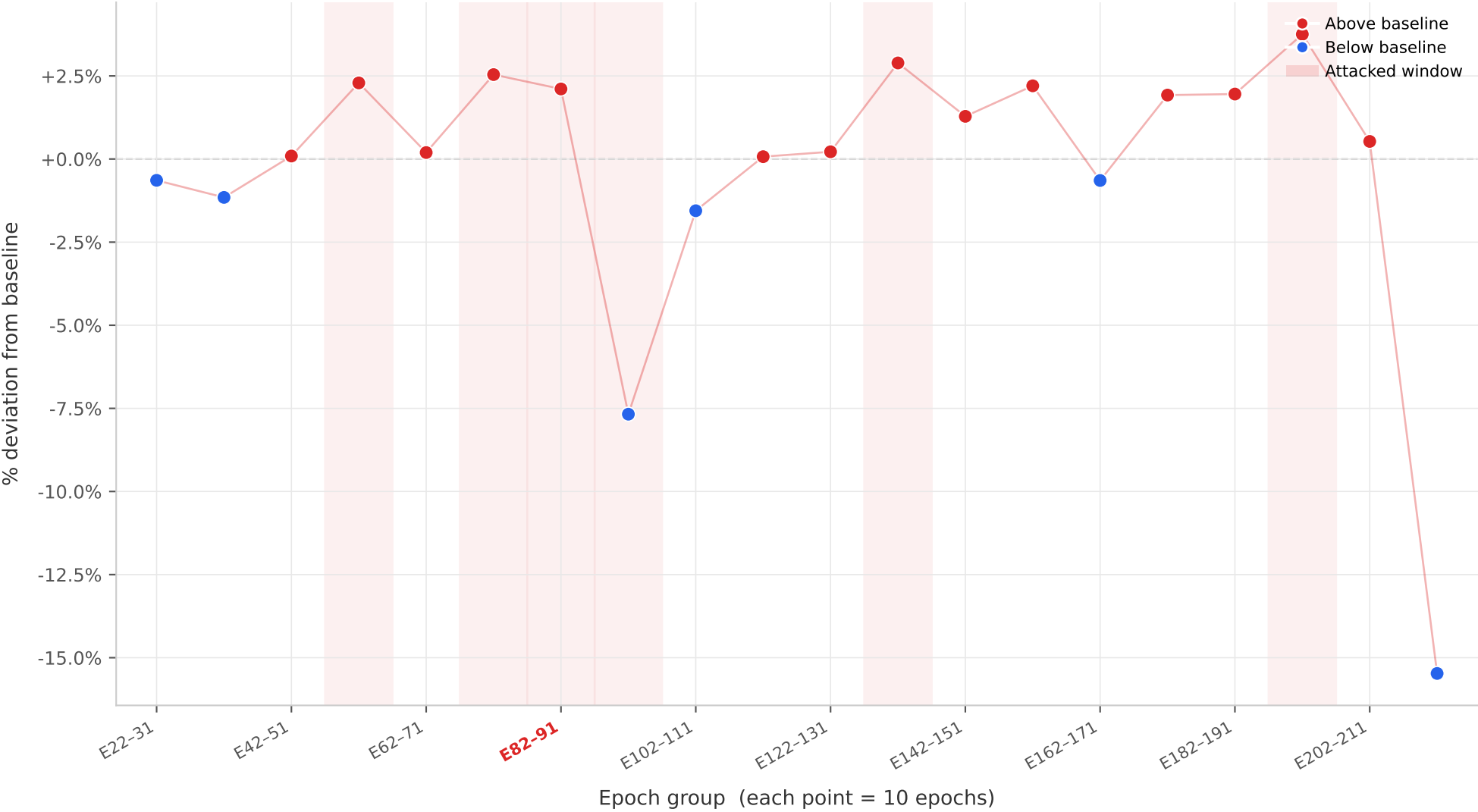
Source IP Cardinality

Cardinality



Destination IP Cardinality

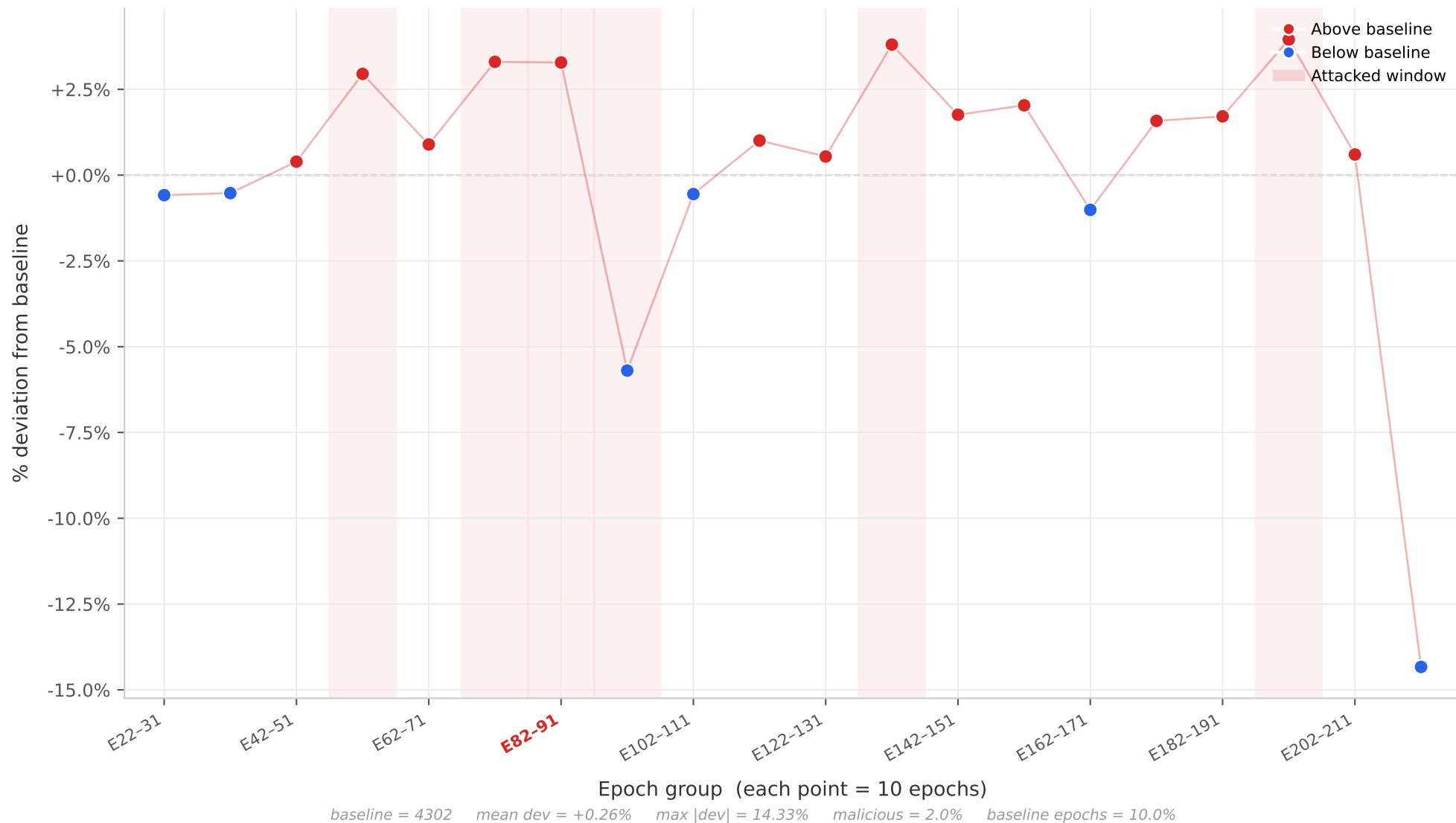
Cardinality



baseline = 5027 mean dev = -0.25% max |dev| = 15.47% malicious = 2.0% baseline epochs = 10.0%

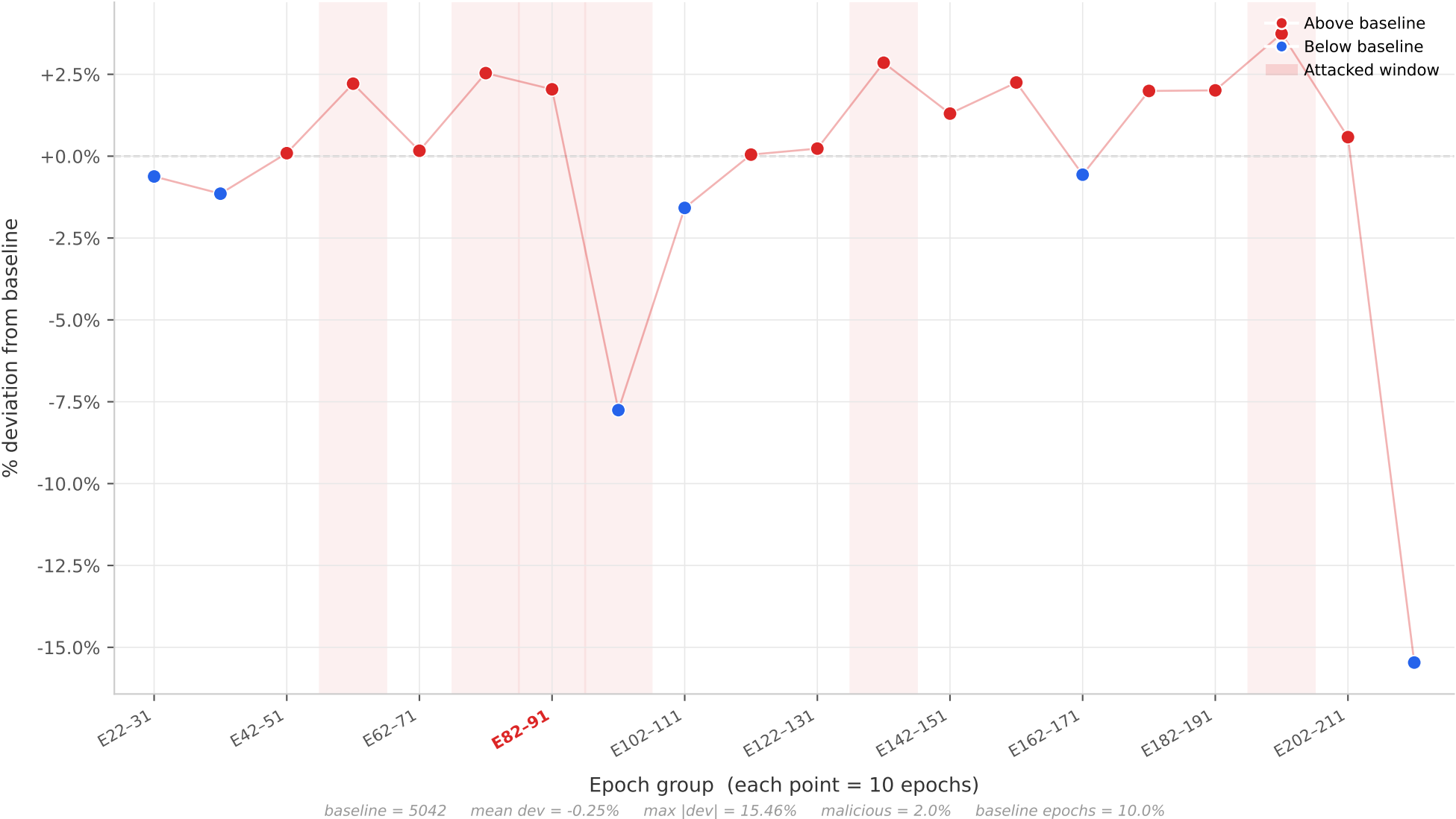
Source Port Cardinality

Cardinality



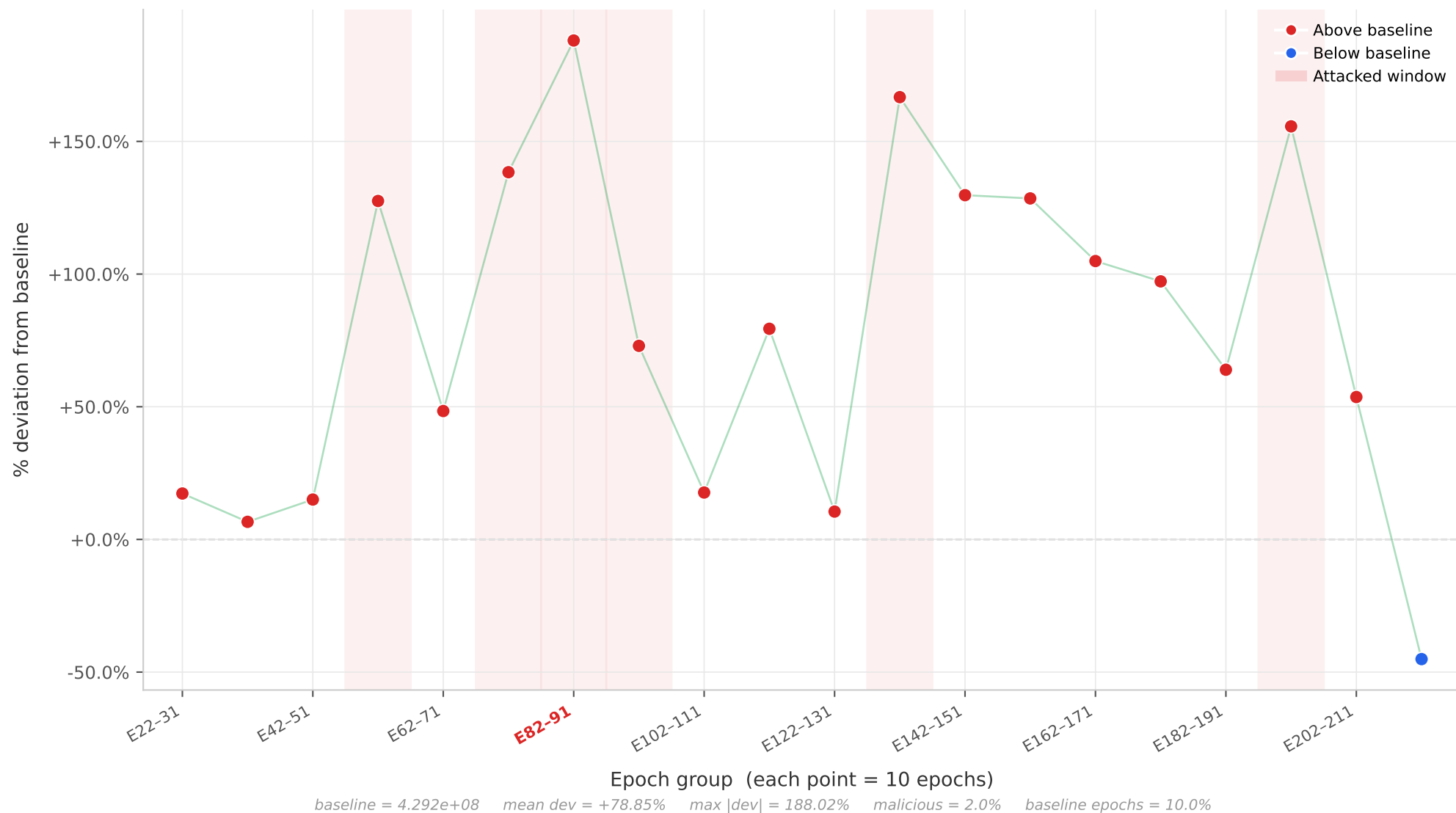
Destination Port Cardinality

Cardinality



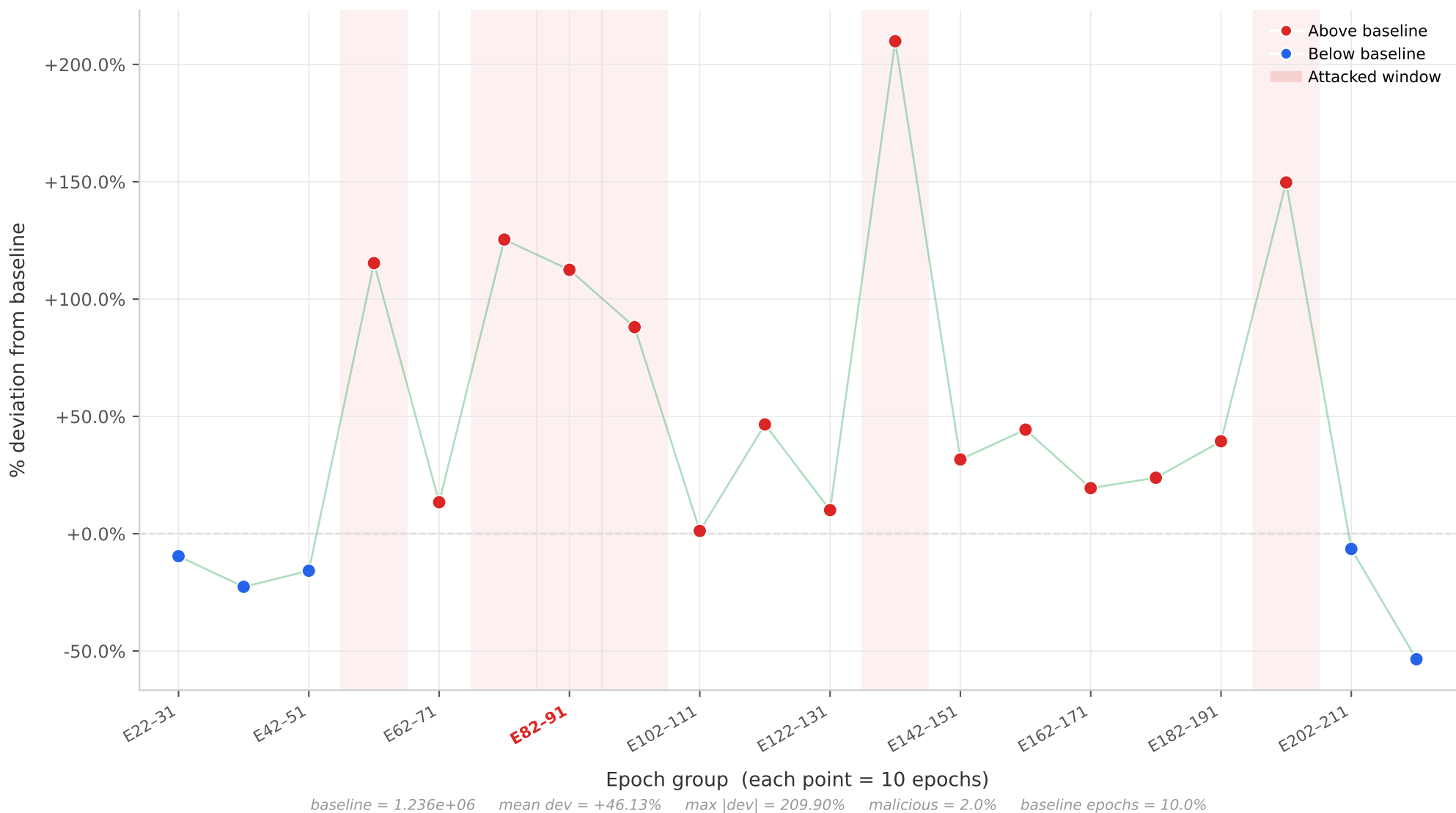
Source IP Uniformity

Uniformity



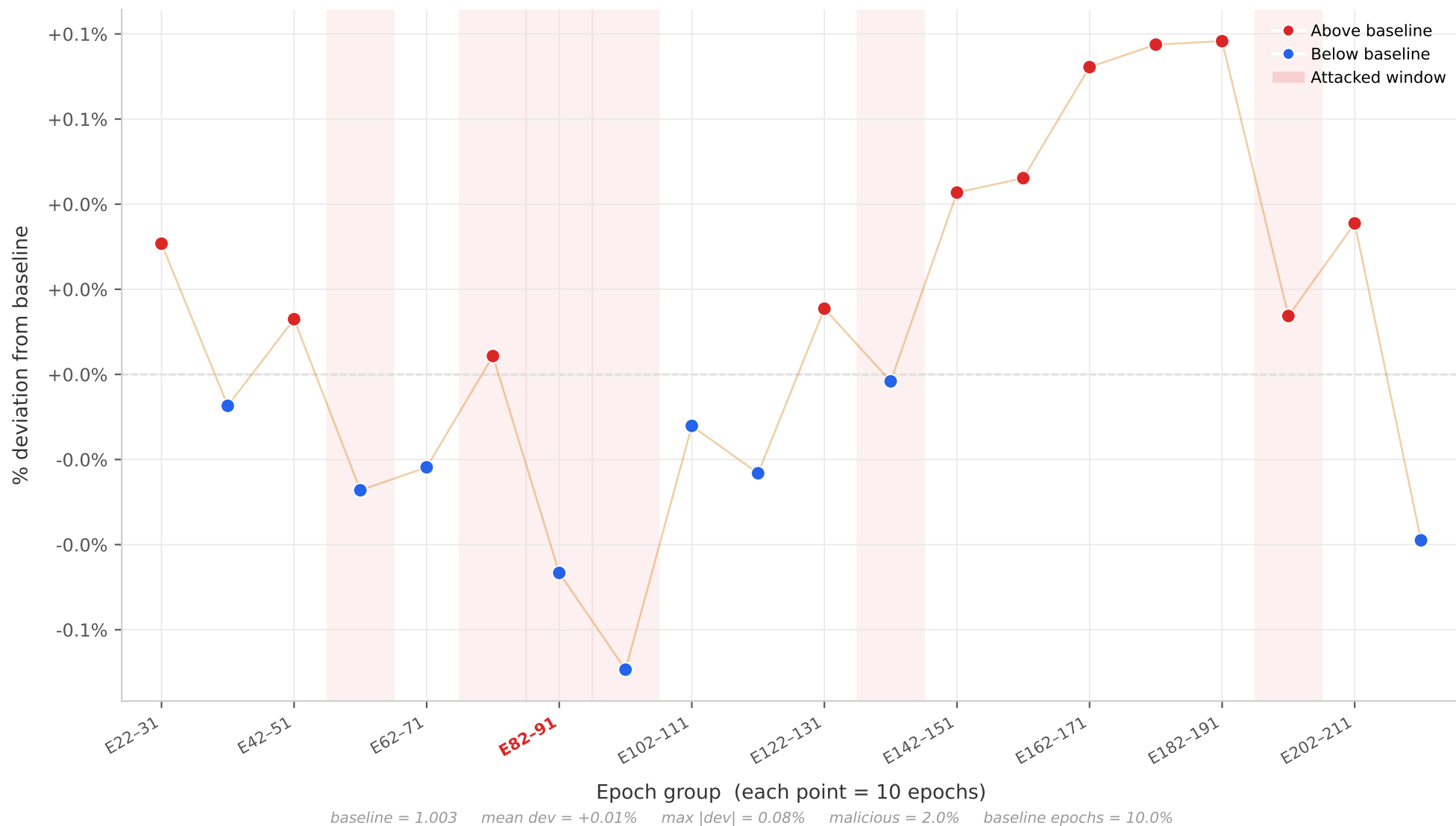
Destination IP Uniformity

Uniformity



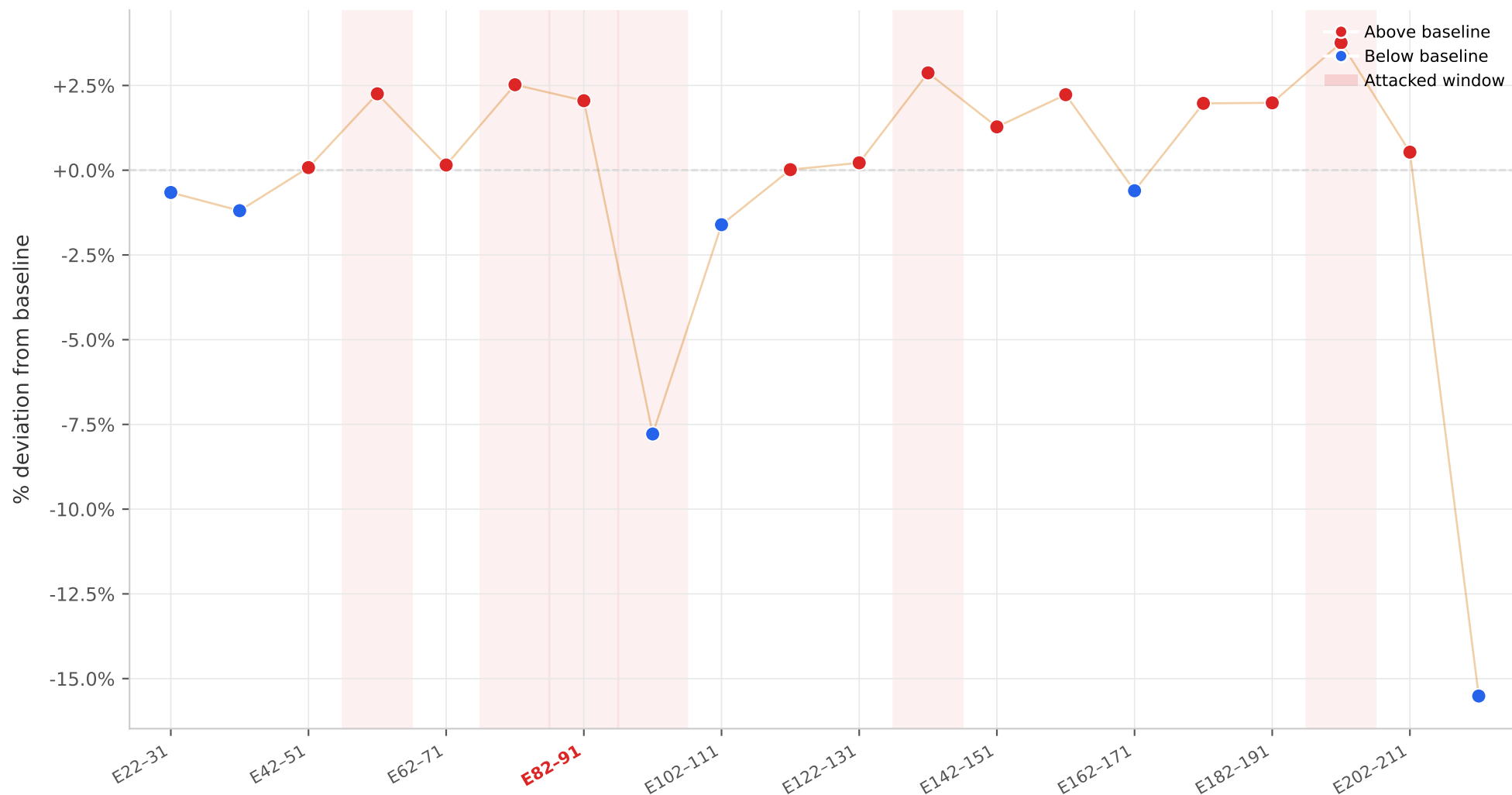
Src / Dst IP Ratio

Flow



Total Flows

Flow

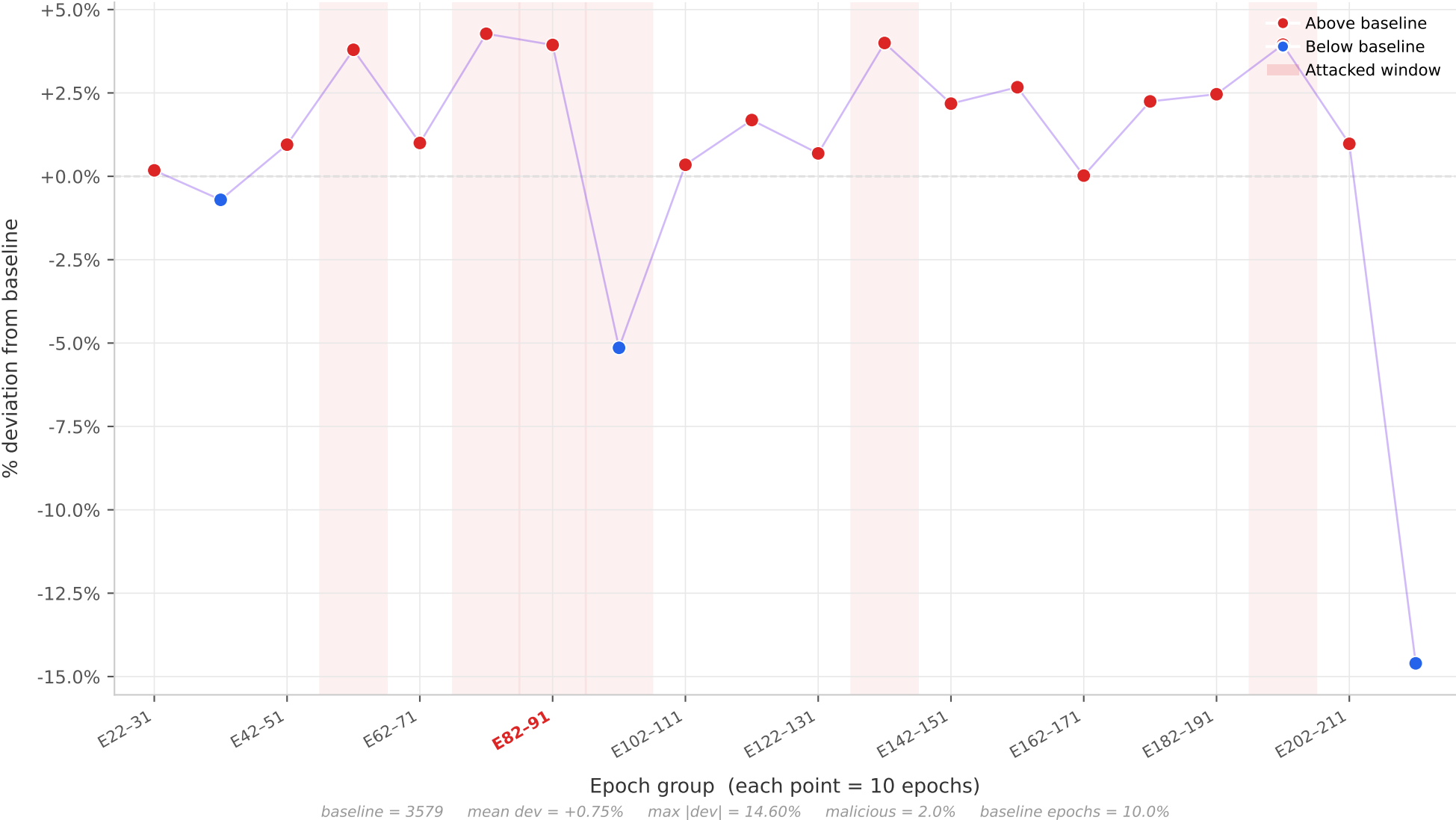


Epoch group (each point = 10 epochs)

baseline = 5078 mean dev = -0.27% max |dev| = 15.51% malicious = 2.0% baseline epochs = 10.0%

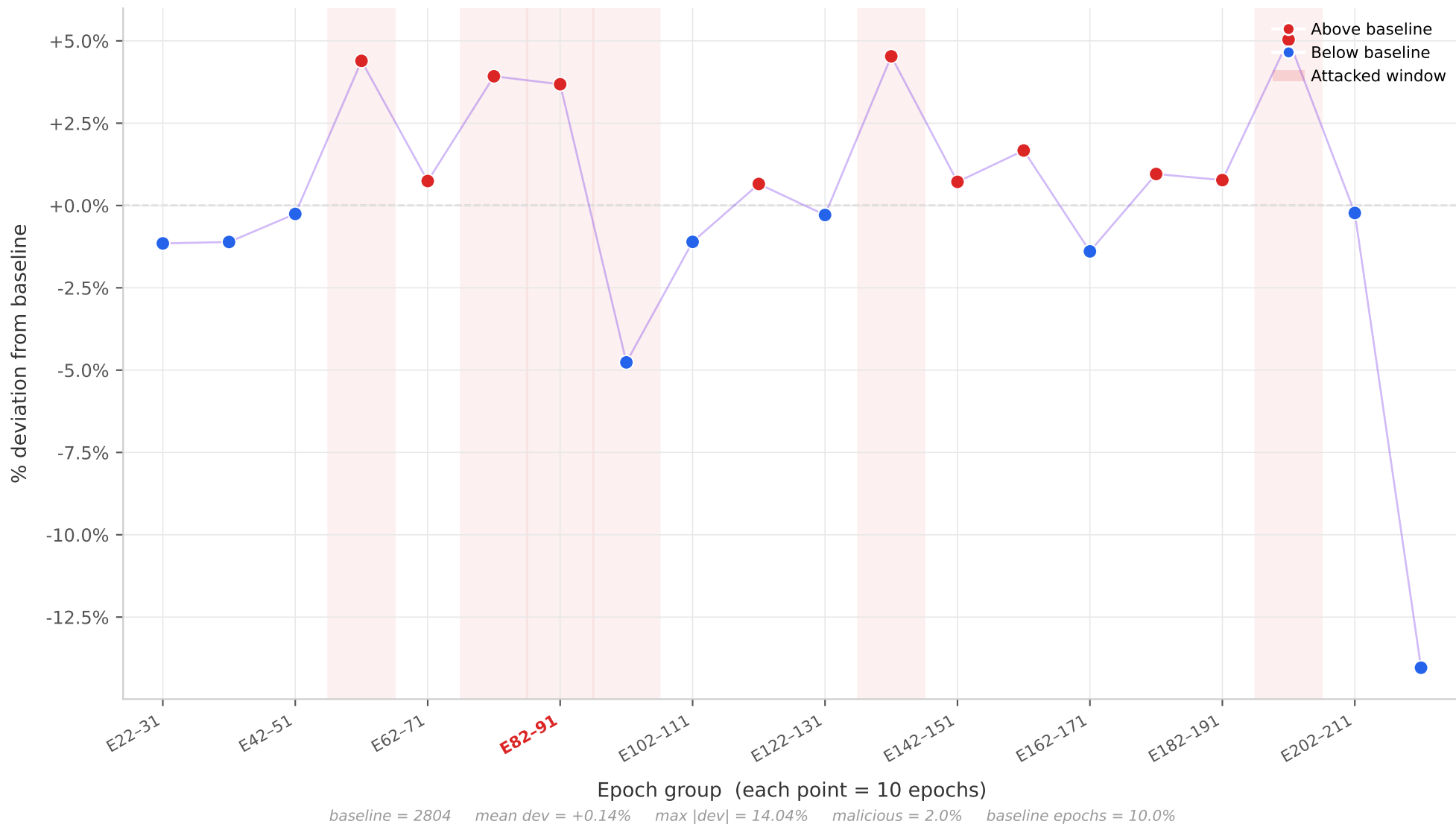
Unique Source IPs

Count



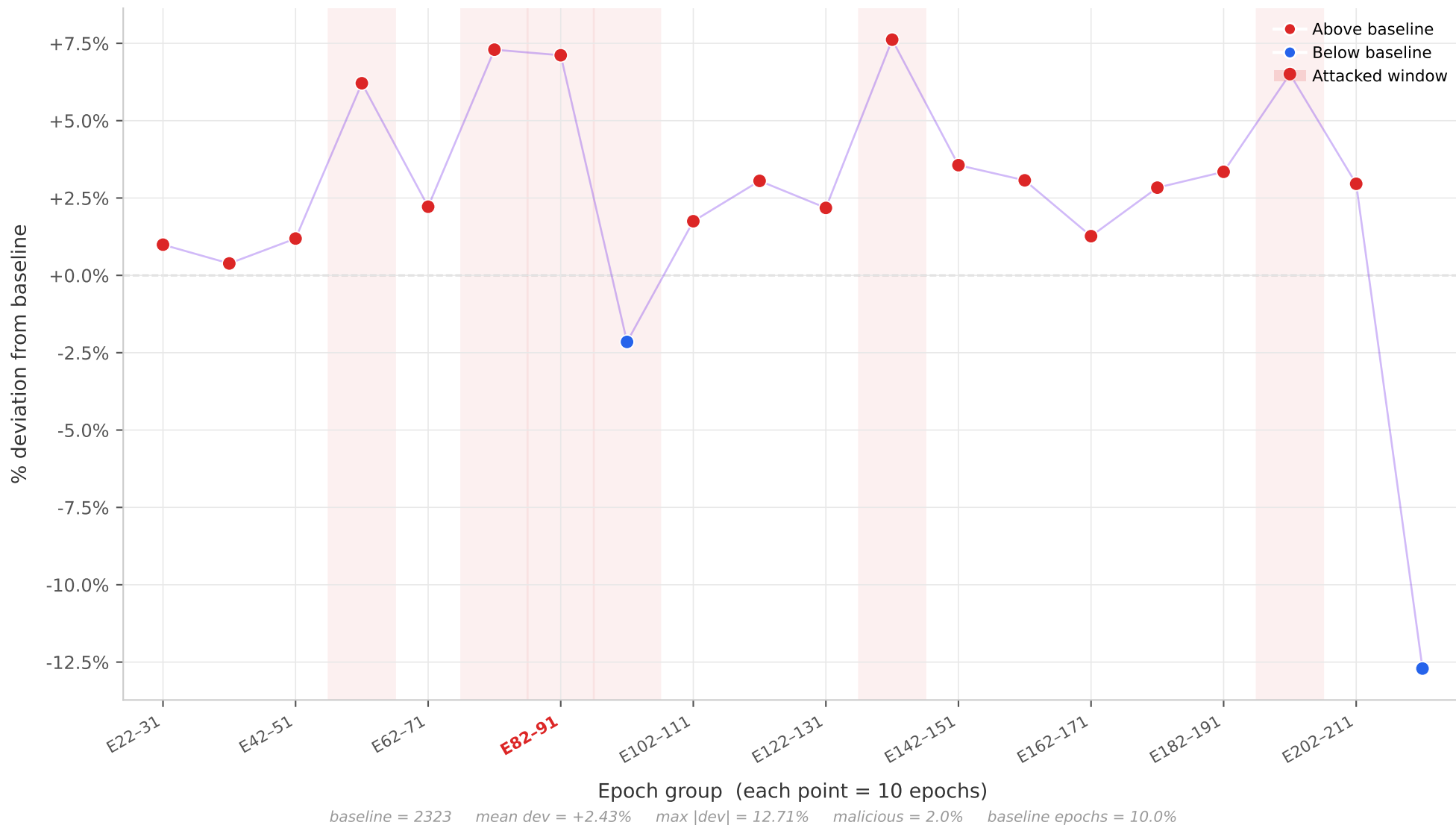
Unique Destination IPs

Count



Unique Source Ports

Count



Unique Destination Ports

Count

